Assignment 4

Multimedia Technologies

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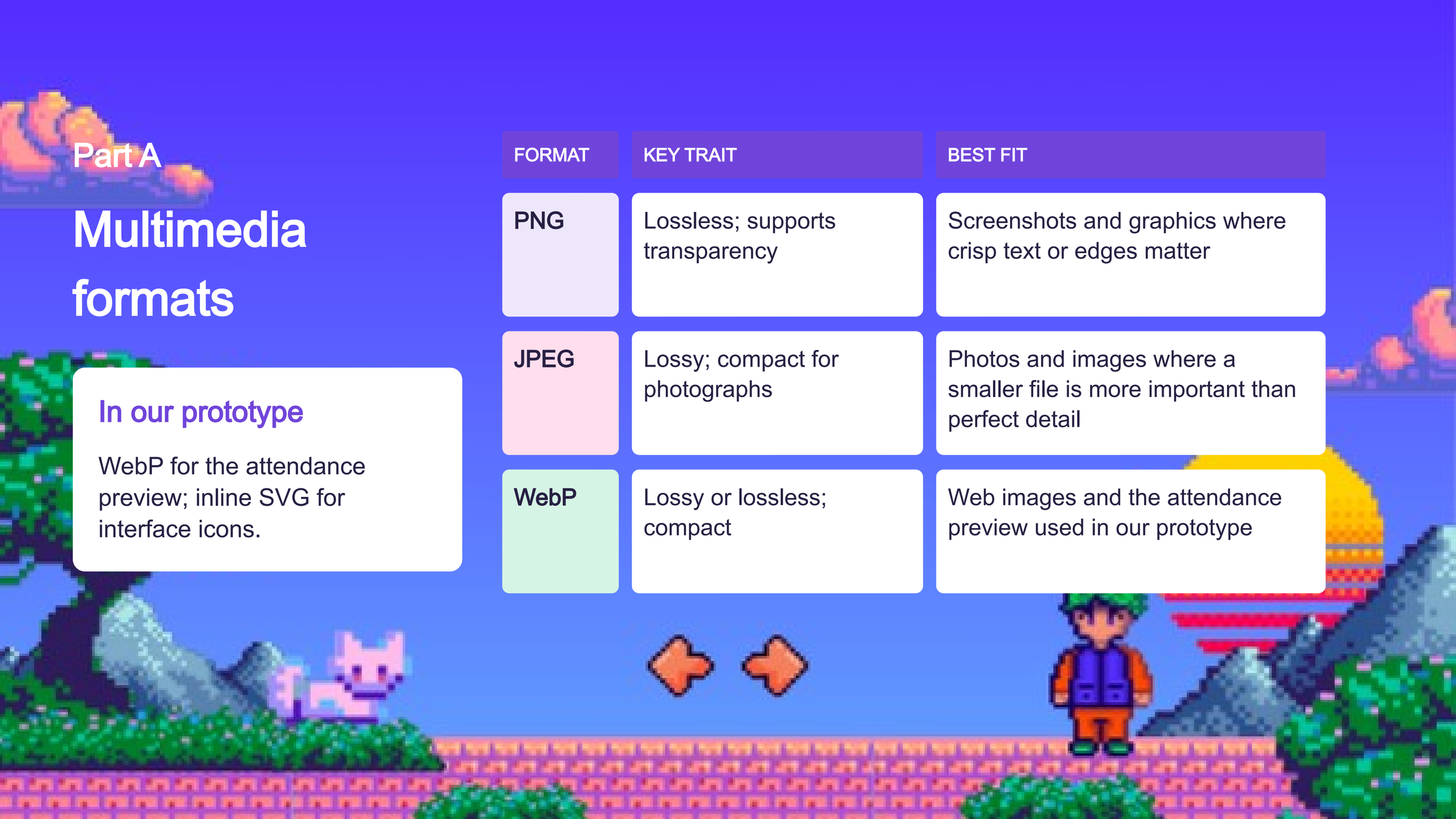
Part A

Multimedia formats

In our prototype

WebP for the attendance preview; inline SVG for interface icons.

FORMAT	KEY TRAIT	BEST FIT
PNG	Lossless; supports transparency	Screenshots and graphics where crisp text or edges matter
JPEG	Lossy; compact for photographs	Photos and images where a smaller file is more important than perfect detail
WebP	Lossy or lossless; compact	Web images and the attendance preview used in our prototype



A1. Format comparison

Format	Type	Compression	Alpha / motion	Typical size	Support	Project use
PNG	Image	Lossless	Alpha; APNG	Large photos	Broad	UI evidence
JPEG	Image	Lossy	Neither	Small photos	Broad	Photos
WebP	Image	Both modes	Alpha + motion	Compact	Modern	App preview
SVG	Image	Vector / text	Alpha + motion	Tiny icons	Broad	UI icons
MP3	Audio	Lossy	N/A	Small	Broad	Speech / music
AAC (M4A)	Audio	Lossy	N/A	Small	Broad*	Narration
WAV (PCM)	Audio	Uncompressed	N/A	Large	Broad*	Editing master
MP4 H.264	Video	Usually lossy	Motion; no alpha	Bitrate-based	Broad*	Demo delivery
WebM VP9	Video	Usually lossy	Motion; alpha*	Bitrate-based	Modern*	Web playback
MOV H.264	Video	Usually lossy	Motion; no alpha	Bitrate-based	Limited web	Editing / transfer

A container (MP4) packages tracks and metadata; a codec (H.264) encodes and decodes the media.

*Codec, OS and device support vary. File size depends on content, resolution, bitrate and duration.
Sources: MDN, Image file type guide and Media container formats.

A2. Measured experiment

PNG · original

58.39 KiB

attendance-preview.png · 390 × 844 px

42.64 KiB saved · 73.0% smaller

WebP Q80 · in app

15.75 KiB

attendance-preview.webp · 390 × 844 px

JPEG exports · pending

attendance-preview.jpg — JPEG Q80

attendance-preview-low.jpg — JPEG Q40

Both: 390 × 844 px. Files not supplied; export and inspect.

10-second clip · pending

Version 1: proposed MP4 H.264, 720p.

Version 2: same clip, lower video bitrate.

Files not supplied; compare text and motion.

Visual quality checks are still pending. JPEG and clip entries are a test plan, not completed measurements.

Source: supplied project inventory, 23 Oct 2026. 1 KiB = 1024 bytes.

A2. Visual quality check

Compare like for like

Use the same attendance screen in all four exports. Compare an identical crop at 100% zoom so text edges and calendar lines remain comparable.

Four exports

PNG

JPEG Q80

JPEG Q40

WebP Q80

Evidence still needed

One side-by-side screenshot, JPEG file sizes, and two exports of the same 10-second clip. Rate each result as sharp, acceptable or poor after inspection.

Comparison images and recording are not included in this deck. Quality ratings remain pending.

A2. Our format choices

For the My AITU / LMS bridge

WebP Q80 supplies the attendance preview at 390 × 844 px. PNG remains the original evidence image because it preserves text and edges without lossy compression.

Inline SVG keeps icons sharp at different sizes and shares the HTML response. CSS provides the login pulse without a separate video download.

Planned demonstration

MP4 with H.264 video and AAC audio for broad playback support. JPEG settings and the ten-second clip still need measured comparison before we can report their quality or size.

73%

smaller preview file

Same pixels. Less data.

PNG 58.39 KiB → WebP 15.75 KiB

42.64 KiB saved at the same resolution.

Measured file sizes only.

Network speed needs its own test.

B1. Interface icons

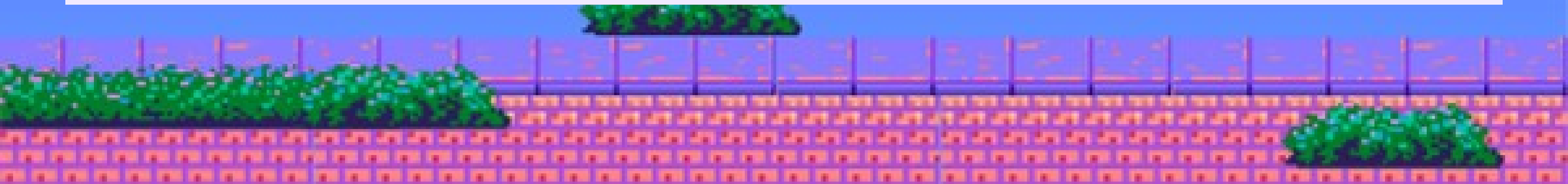
#	Asset	Where used	Vector viewBox	Markup size
3	**Book**	Courses / navigation	SVG 300 × 300	740 B
4	**Calendar**	Tasks / attendance	SVG 300 × 300	1,162 B
5	**Graduation**	Grades	SVG 300 × 300	857 B
6	**User**	Account button	SVG 300 × 300	869 B
7	**Menu**	Navigation button	SVG 300 × 300	561 B
8	**Close**	Navigation controls	SVG 300 × 300	441 B
9	**Down**	Account menu	SVG 300 × 300	391 B
10	**Sign-out**	Account menu	SVG 300 × 300	906 B
11	**Refresh**	Reload	SVG 300 × 300	945 B
12	**Search**	Course search	SVG 300 × 300	669 B
13	**Eye**	Password visibility	SVG 300 × 300	935 B
14	**Check**	Attendance	SVG 300 × 300	366 B

All icons: Flaticon Uicons Regular Rounded. Free licence with visible footer attribution.
Inline symbols share the HTML response; these are source sizes, not separate requests.

B1. Media asset inventory

#	Asset	Where used	Format / dimensions	Size	Source / licence
15	**Absent icon**	Attendance	SVG 300 × 300	442 B	Flaticon[1]
16	**Clock icon**	Attendance time	SVG 300 × 300	777 B	Flaticon[1]
17	**Minus icon**	Unmarked attendance	SVG 300 × 300	284 B	Flaticon[1]
18	**Left arrow icon**	Previous month	SVG 300 × 300	416 B	Flaticon[1]
19	**Next arrow icon**	Next month	SVG 300 × 300	404 B	Flaticon[1]
20	**List icon**	Attendance status	SVG 300 × 300	1,010 B	Flaticon[1]
1	**Attendance preview**	Sign-in	WebP 390 × 844	15.75 KiB	Own + icons[1]
2	**LMS wordmark**	Sign-in	CSS / text, 68 × 68	Embedded	Own
21	**Login pulse**	Login progress	CSS, 2 s loop	Embedded	Own
22	**Entrance motion**	Transitions	CSS, 0.4 s	Embedded	Own
23	**Campus map**	Map section	Responsive iframe	Variable	Yuujiso / MIT[2]
24	**Inter font**	Interface text	Web font / weights	Variable	Inter / OFL[3]
25	**Remote view**	Optional support	JPEG Q70, 1100 × 800	Variable	Private session

[1] Flaticon attribution in footer. [2] github.com/Yuujiso/aitumap (MIT), author beside map.
[3] github.com/rsms/inter (SIL OFL 1.1). Remote session views must stay out of the submitted demo.
Unused ui-right / ui-stats and evidence PNG files are not loaded by the running interface.



B2. Integration and optimisation

What is integrated

WebP preview: 390 × 844 px, displayed at 250 px wide.

18 inline SVG icons and a CSS login pulse.

Reduced-motion preference disables animation.

The full page still needs a 360 px browser check.

−73.0%

Preview image

59,787 B → 16,126 B

−30.8%

4 front-end files + preview

141,601 B → 97,941 B

DevTools measurements: still required

Disable cache, select Fast 4G and reload the same screen before and after. Capture transferred bytes, request count and Finish/Load time. Image-request delta should be 0; verify the total.

Controlled local payload excludes headers, compression, API, fonts and map traffic. Stage 1: <3 s per request at 1,000 users remains unverified.

B3. Accessibility of media

Meaningful images have alt text

The test attendance preview describes the month and status counts. Decorative SVGs use aria-hidden="true"; controls retain text or accessible labels.

```
<img src="/attendance-preview.webp"
width="390" height="844"
alt="Тестовый экран посещаемости: календарь за октябрь 2026 года и счётчики — 0 присутствий, 1 пропуск, 0 опозданий, 1 запись без отметки.">
```

Motion follows user preference

Progress is visible as text and a pulse. Animation and transitions are disabled when reduced motion is requested.

```
@media (prefers-reduced-motion: reduce) {
  *, *::before, *::after { animation: none !important;
  transition: none !important;
  scroll-behavior: auto !important; } }
```

No speech media or sound playback: captions and sound equivalents are currently N/A. No live text sits over raster images, so the 4.5:1 text-over-image test is N/A. Source: supplied part_b.md, B3.

Part C — Multimedia presentation



C1. Storyboard

01 · 0:00–0:30

Login screen

Nurassyl: how the login process looks

02 · 0:30–1:10

Main pages, burger menu and references from Stage 1

Salima: website content and design decisions

03 · 1:10–1:40

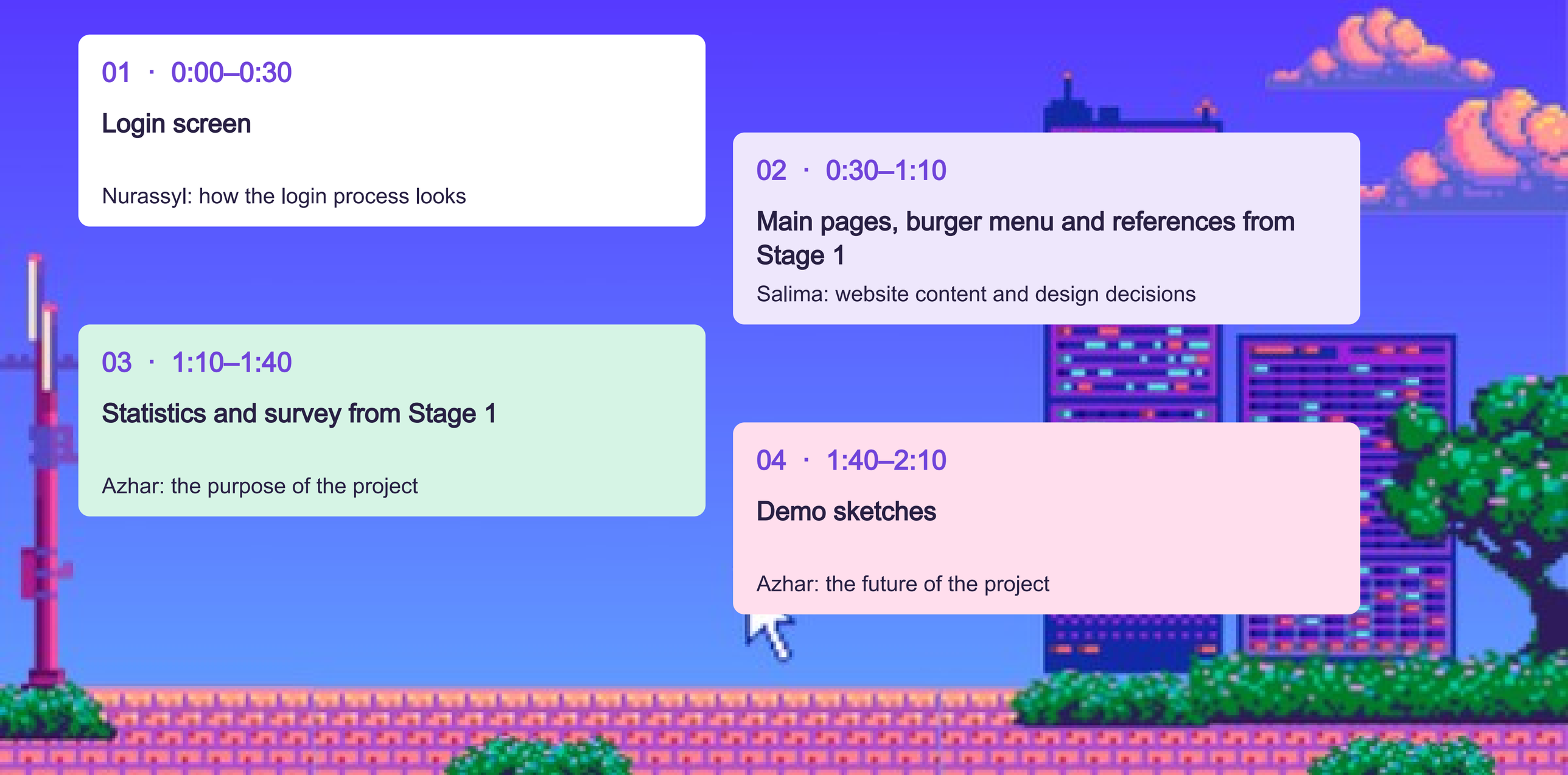
Statistics and survey from Stage 1

Azhar: the purpose of the project

04 · 1:40–2:10

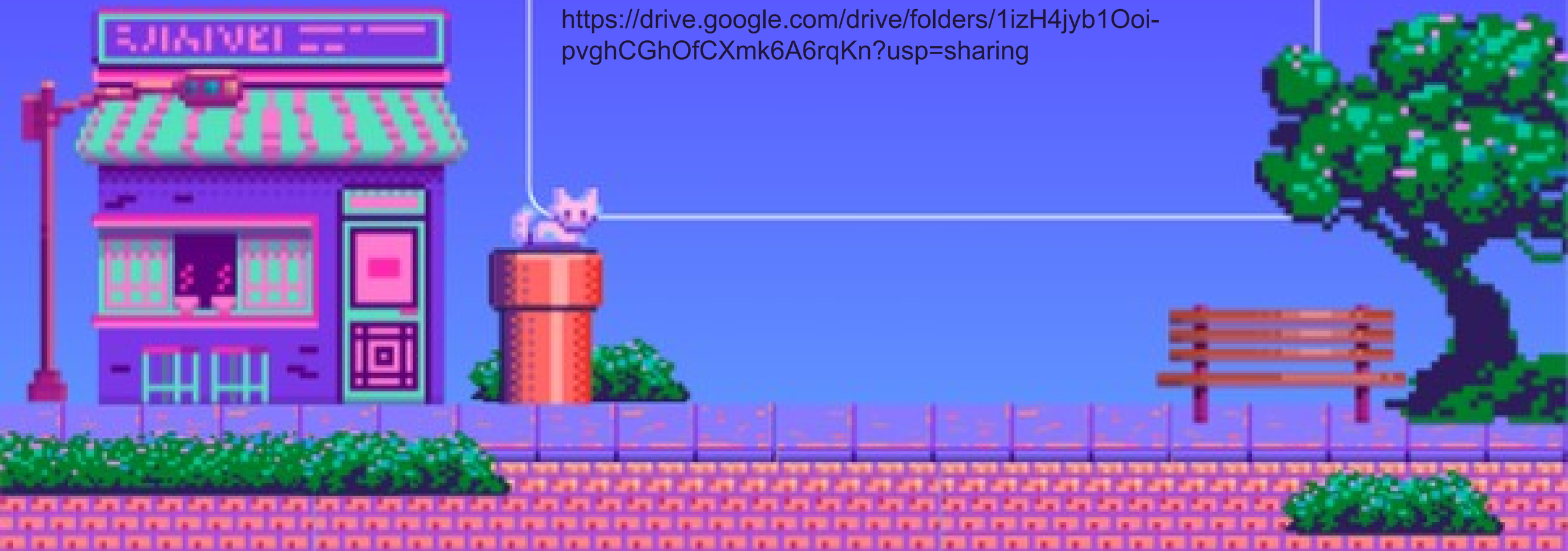
Demo sketches

Azhar: the future of the project



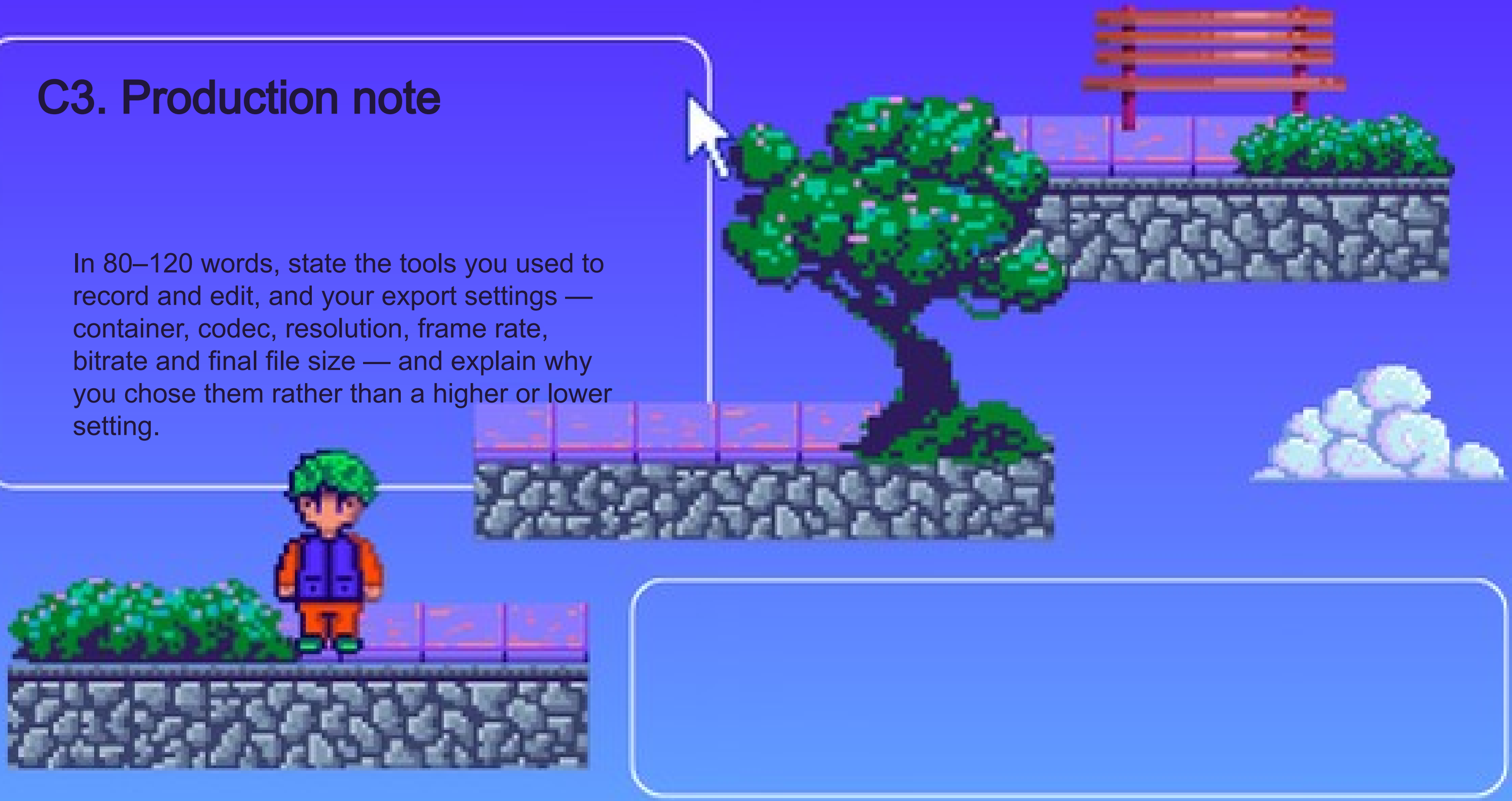
C2. The video

https://drive.google.com/drive/folders/1izH4jyb1Ooi-pvghCGhOfCXmk6A6rqKn?usp=sharing



C3. Production note

In 80–120 words, state the tools you used to record and edit, and your export settings — container, codec, resolution, frame rate, bitrate and final file size — and explain why you chose them rather than a higher or lower setting.



Part D — Integration with the project



D1. Updated prototype

Run the prototype

github.com/d30mantcoder/ICTProject

01 Clone the repository

02 Create a virtual environment

03 Install requirements.txt

04 Run bridge-test/app.py with Python



D2. Change log

01 Real LMS data

Added fetching from LMS to make the website usable.

Status: using real LMS data.

02 A more usable interface

The original LMS UX was uncomfortable. The redesigned interface places the most useful buttons on the main page.

03 Multi-user sessions

Previously, more than one person could not work at the same time. Asynchronous sessions now let the server handle multiple users.

Data → interface → concurrent use



D3. Team contributions

C, D

Salima

CONTRIBUTION

Editing the video

A, D

Azhar

CONTRIBUTION

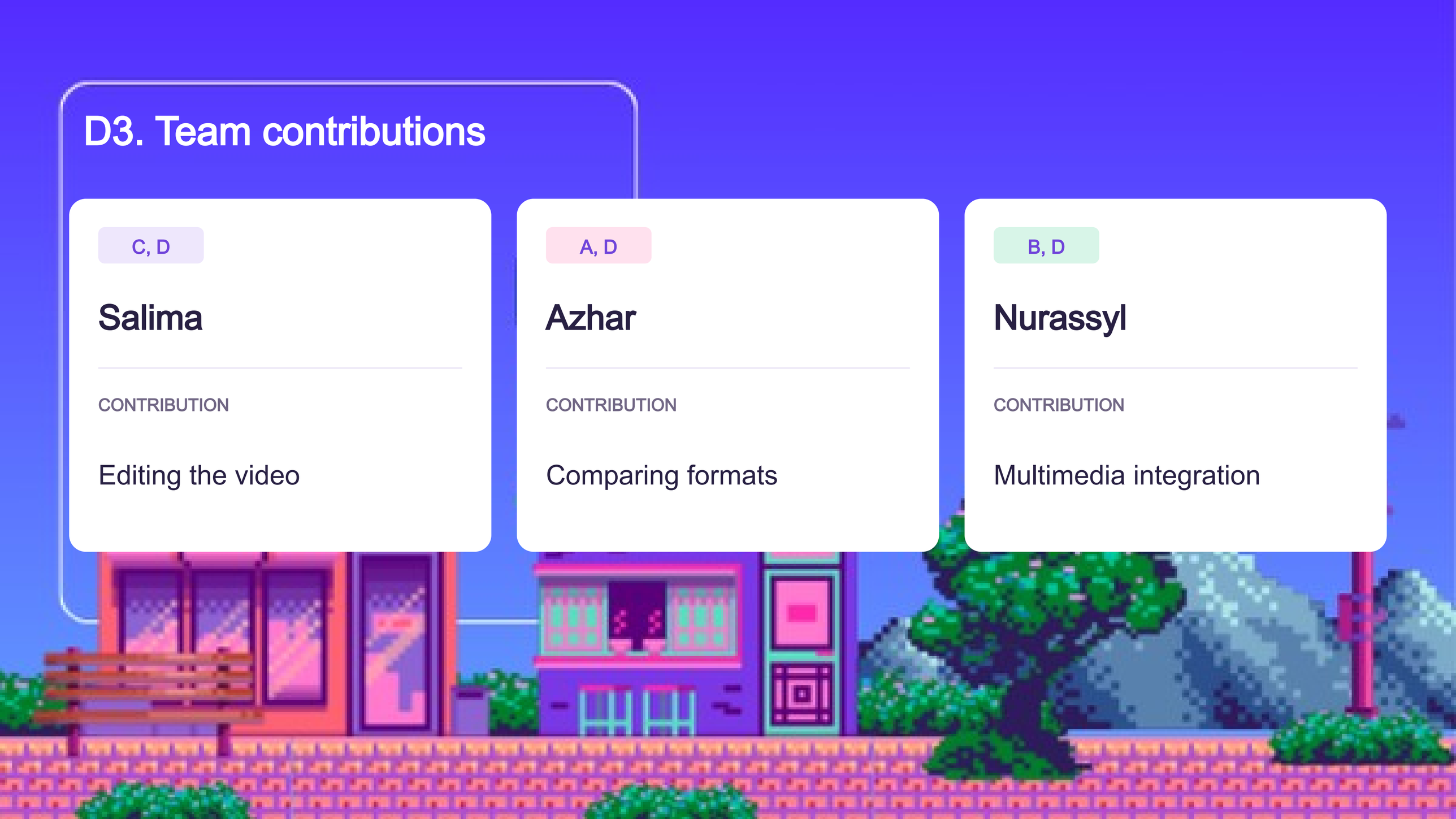
Comparing formats

B, D

Nurassyl

CONTRIBUTION

Multimedia integration



Thank You

